

EXP2-4 Part 2 Report M18

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1. Introduction

To determine the factors that affect the strength of the Whole Tooth Company dental adhesive, the experimental team conducted the following factorial design experiments. The individual factors are polymer type, water-to-solid ratio, solidification temperature, and type of glass powder. Each factor is labeled as two different levels to determine the factors that affect the strength of the final filling material, as shown in Table 1, in which.

- ◆ P1 is an old polymer, and P2 is a new type of polymer.
- ◆ Two ratios of water to solid, 1:7 and 1:9, were selected to test the effect of the amount of water used.
- ◆ The mixture was solidified and stored at two different temperatures, 35 °C, and 20 °C, for 24 hours until testing.
- ◆ Glass powder from two suppliers, G1 is old glass powder and G2 is new glass powder.

Table 1. Factors and Level Used in the Factorial Designed Experiment

Factors		Levels	
		+	-
A	Polymer	P1	P2
B	Water:Solid	1:7	1:9
C	Temperature	35°C	20°C
D	Glass Powder	G1	G2

The experimental group designed factor experiments and conducted compressive strength tests on all the samples to obtain compressive strength data under the influence of each factor. According to the factorial experimental design, 16 runs were conducted five repeated experiments were conducted, and the results were analyzed.

2. Results of Calculation

According to the calculation method[1], all the data are calculated as below.

Table 2. Factors, Interaction Design and Outcomes

Run Order	What Have The Effect										Outcome					Calculations	
	Factors				Interaction						Compressive Strength/Mpa					Y'	R
	A	B	C	D	AB	AC	AD	BC	BD	CD	1	2	3	4	5	Mean/MPa	Range/MPa
Run 1	+	+	+	+	+	+	+	+	+	+	43.4	28	22.2	43.6	34.3	34.3	21.4
Run 2	-	+	+	+	-	-	-	+	+	+	39.2	22.6	24.3	92.8	45.2	44.82	70.2
Run 3	+	-	+	+	-	+	+	-	-	+	7.1	47.4	15	9.9	19.8	19.84	40.3
Run 4	-	-	+	+	+	-	-	-	-	+	75.5	67.1	41.9	48	76.8	61.86	34.9
Run 5	+	+	-	+	+	-	+	-	+	-	15	15.8	14.7	14.9	15.3	15.14	1.1
Run 6	-	+	-	+	-	+	-	-	+	-	52.1	30.1	79.8	35.2	32	45.84	49.7
Run 7	+	-	-	+	-	-	+	+	-	-	7.1	2.1	9.2	9.3	6.9	6.92	7.2
Run 8	-	-	-	+	+	+	-	+	-	-	112.3	118	31.7	157.5	145.7	113.04	125.8
Run 9	+	+	+	-	+	+	-	+	-	-	41.6	32.8	21.9	23.6	28.3	29.64	19.7
Run 10	-	+	+	-	-	-	+	+	-	-	41.8	23.9	19.4	37.4	78.4	40.18	59
Run 11	+	-	+	-	-	+	-	-	+	-	38.3	17.8	16.6	29.6	24.4	25.34	21.7
Run 12	-	-	+	-	+	-	-	-	+	-	76.4	65.8	49.2	41.7	55.5	57.72	34.7
Run 13	+	+	-	-	+	-	-	-	-	+	15.3	16.1	14.7	13.8	14.9	14.96	2.3
Run 14	-	+	-	-	-	+	+	-	-	+	48.1	67.5	32.4	36.3	28.5	42.56	39
Run 15	+	-	-	-	-	-	-	+	+	+	6.9	3.4	10.1	8.7	7.3	7.28	6.7
Run 16	-	-	-	-	+	+	+	+	+	+	89.2	113	135.5	142.6	58.2	107.7	84.4

By using the following formula, the calculation results can be obtained as shown in Table 3 and Table 4.

$$SUM + (Y') = \sum Y'_+ \quad (1)$$

$$SUM - (Y') = \sum Y'_- \quad (2)$$

$$SUM + (R) = \sum R_+ \quad (3)$$

$$SUM - (R) = \sum R_- \quad (4)$$

$$Effect = \frac{|(SUM+) - (SUM-)|}{N} \quad (5)$$

$$Random\ Variation = \frac{Average\ R}{N} \quad (6)$$

Table 3. Mean Values and Random Variation

Results	Value	Unit
Sum Mean Compressive Strength	667.14	MPa
Mean Range	38.63125	MPa
N	8	/
Random Variation	9.6578125	MPa

Table 4. Effect for Outcomes Y' and Range

Factors	Mean Outcome Y' /MPa				Range R/MPa			
	SUM+	SUM-	Diff	Effect	SUM+	SUM-	Diff	Effect
A	153.42	513.72	360.30	45.04	120.40	497.70	377.30	47.16
B	267.44	399.70	132.26	16.53	262.40	355.70	93.30	11.66
C	313.70	353.44	39.74	4.97	301.90	316.20	14.30	1.79
D	341.76	325.38	16.38	2.05	350.60	267.50	83.10	10.39
AB	434.36	232.78	201.58	25.20	324.30	293.80	30.50	3.81
AC	418.26	248.88	169.38	21.17	402.00	216.10	185.90	23.24
AD	324.36	342.78	18.42	2.30	287.10	331.00	43.90	5.49
BC	383.88	283.26	100.62	12.58	394.40	223.70	170.70	21.34
BD	338.14	329.00	9.14	1.14	289.90	328.20	38.30	4.79
CD	333.32	333.82	0.50	0.06	299.20	318.90	19.70	2.46

Based on the comparison of the size of Effect and Random Variation, the factors that have been **preliminarily determined** in the experiment to affect the performance or of the material are as follows:

Factors and Interactions that Significantly Affect Performance	Factors and Interactions that Significantly Affect Process Reliability
Type of Polymer	Type of Polymer
The ratio of water to solid	The interaction between polymer and temperature
The interaction between polymer and water	The interaction between water and temperature
The interaction between polymer and temperature	

3. Factors and Interaction Analysis

The factors or interactive factors that lead to product degradation can affect material performance or process reliability. To consider the impact of these two characteristics, the results (Y) and range (R) are used as quantitative indicators of performance and reliability. If the impact of the results or range is greater than the random error of the testing method, then it does indeed have an impact on the corresponding characteristics. If the effect value of each factor or interaction is less than the change, it can be considered that it does not affect the characteristic, otherwise, it will affect.

■ Influence on Material's Performance

SUM+(Y') and Y SUM-(Y') are the result that represents material performance in different cases. If the Y effect is larger than the random variation, the factors or interactions influence the product's performance and this influence could be analyzed directly by the outcomes. Therefore, it is necessary to calculate and compare them to determine which elements or types of influencing factors have a greater impact on product performance.

By comparing the effects with random errors, it can be preliminarily determined that:

When considering the influence of single factors:

Both polymer and water solid ratios have a significant impact on the overall performance

When considering the interaction between two factors:

In the case of polymer with water solid ratio, both factors have an impact on product performance and affect each other's performance in the interaction.

In the case of polymer with temperature, the polymer will have an impact on how temperature will behave

■ Influence on Process Reliability

SUM+(R) and SUM-(R) represent the process reliability in different cases. Therefore, it is necessary to calculate and compare them to determine which elements or types of influencing factors have a greater impact on process reliability.

By comparing the calculation results with random errors, it can be preliminarily determined that:

When considering the influence of single factors:

Polymers have a significant impact on performance fluctuations and have a significant impact on the reliability of the process.

When considering the interaction between two factors:

In the case of polymers with temperature, the influence of polymers is more significant, while the influence of temperature can be ignored. Meanwhile, polymers will have an impact on temperature-induced behavior, leading to changes in process reliability.

In the case of water solid ratio with temperature, although the individual effects of the two factors are not significant, their interaction has a significant impact on the reliability of the process

■ Total Influence on Materials

In summary, from the above analysis, it can be seen that some factors or interactions may have an impact on the compressive strength or R of the product.

For Single Factor:

On the one hand, the use of new polymers significantly improves compressive strength, but it also leads to a decrease in the reliability of the material manufacturing process, which sometimes results in poor performance of products using new polymers. The two different polymers, old and new, undergo different chemical reactions under the same conditions, resulting in these effects. In addition, different polymers also imply differences in swelling volume[2], which may mislead dentists and add excessive water during actual use.

On the other hand, the lower water-to-solid ratio will lead to a decrease in overall compressive strength. Water affects the release of Al^{3+} , Ca^{2+} and F^{-} , which promote the reaction.[3] However, excessive water can also lead to insufficient ion concentration. Due to insufficient ion concentration in the system, the reactants cannot fully contact and react, resulting in a slower reaction rate and inability to achieve the theoretical complete conversion rate. The crosslinking degree of the product is low, ultimately leading to a decrease in compression strength. It might be the main reason for the poor performance of the new product.

For Interactions between Factors:

The interaction between polymer type and curing temperature also affects compressive strength. Based on other data, it can be analyzed that polymers have an impact on temperature, which is due to changes in the polymer causing changes in the reaction itself. This is not only reflected in the changes in compressive strength but also leads to a decrease in material reliability.

The interaction between polymer type and water solid ratio also affects compressive strength. The type of polymer has an impact on the strength change caused by the water-solid ratio, and the water-solid ratio also has an impact on the compressive strength of the polymer type. The interaction between the two leads to changes in compressive strength.

In addition, the interaction between water solid ratio and solidification temperature also leads to issues with material reliability, which can also randomly lead to a decrease in performance. This is due to the influence of the water-solid ratio in the combination.

4. Blamestorming

During the blamestorming conversation, the team's practical observations largely align with the findings from the calculations.

Simon and Anita correctly noted that New Polymer has better properties, which matches the calculated increase in strength. However, the calculation also highlights that the new polymer's decreased process reliability could lead to random deterioration, an aspect that might have been underappreciated. Simon's assessment of temperature was insightful: while he correctly identified that temperature indirectly affects performance through its interaction with other factors, the varied impacts on different combinations with temperature suggest more nuanced influences than initially discussed.

Laura and Darren's experience with the new polymer's stickiness led Darren to add more water, but the calculations reveal that a higher water-solid ratio with the new polymer reduces performance, indicating a practical misinterpretation. Laura's opinion on glass powder has been proved by data, and this will not affect the reaction process and material strength.

The company has adopted a trusting attitude towards suppliers and has not conducted experiments to verify whether there are differences in glass powder. However, although the experimental results were not reflected, there may still be potential impacts. On the other hand, the company also has serious deficiencies in batch management of production materials and understanding of material manufacturing processes and parameters. The company urgently needs to address these issues.

Overall, the team's insights are mostly correct but must carefully explain the experimental results to avoid errors and achieve consistent product quality. Actual observation and calculation of data can guide better decision-making and process adjustments in dental cement manufacturing.

5. Critical Thinking for Original Experimental Plan

The experimental plan designed by the group adopted the method of controlling variables. The other elements in the scene remain consistent when testing another experimental element. Compared with the reference experiment, the experimental plan designed for this project only conducted preliminary tests on the experimental environment at 21 °C and 35 °C, lacking further testing on the impact of other variables on the reaction under 21 °C conditions. In addition, the group has set a more detailed classification for the experiment on water solid ratio, which can help the company better understand the impact of changes in water solid ratio on materials.

On the other hand, compared with the reference experiment, the experimental plan designed in this project overlooks the interaction between multiple factors, and the interaction between each factor cannot be fully reflected in our experimental plan.

At the same time, there is a lack of consideration for the placement time before the reaction begins, which may lead to defects in the material being tested without fully curing.

6. Future Plan Suggestions for the Company

As a company, in order to control the losses caused by product performance defects, affected products should be temporarily suspended in a timely manner to prevent the situation from escalating and causing greater losses. At the same time, further improvements should be made to the product, identifying the specific reasons that affect the filling materials and making corrections. At the same time, the company should carefully monitor the production process to improve the reliability of product quality.

From the experimental conclusion, it can be concluded that polymer type and water-solid ratio are the main influencing factors, while temperature is a physical quantity that affects the reaction environment. Through the analysis of experimental results, it can be concluded that the selected polymer and water solid ratio are the main factors affecting the compressive strength of dental fillers. In addition, when the polymer-to-water solid ratio is appropriate, the crosslinking temperature will also have an impact. Based on the above statement, it is recommended to conduct further experiments to determine the optimal conditions for tooth filling: Given the impact of new polymers on material performance improvement, using new polymers for factor design experiments is more reasonable. Meanwhile, the experimental process should be supervised to improve reliability. It is recommended to further refine the water-solid ratio and conduct factorial experiments to determine the optimal combination of new polymers and the water-solid ratio. Considering that the oral temperature, it is recommended to improve the temperature control and monitoring during the testing period so that the above experiments can be conducted in an

environment of 35 °C.

After confirming the final water-to-solid ratio, it is recommended that the company provide recommendations to the dentist on the water-to-solid ratio to curb the impact of excessive water volume caused by subjective judgment.

Reference

- [1] Xinru. Deng. EXP2-4 Introduction Part 2-23B
- [2] H. Omidian, S.- . Hasherni, F. skari, and S. Nafisi, 'Swelling and rosslink ensit Measurements for H drogels', Iran. J. Polym. Sci. Technol., vol. 3, Jun. 1994.
- [3] Xinru. Deng. EXP2-4 Introduction Part 1-23B